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Deep Learning Cheat Sheet



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Deep Learning Basics

Neural Networks –

Composed of input, hidden, and output layers

Activation Functions –

Sigmoid, ReLU, Tanh, Softmax

Loss Functions – MSE, Cross-Entropy

Optimizers – SGD, Adam, RMSprop



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Simple Neural Network

Feedforward Neural Network (FNN)



```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(64, activation='relu',
          input_shape=(10,)),
    Dense(32, activation='relu'),
    Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam',
              loss='binary_crossentropy', metrics=
              ['accuracy'])
model.summary()
```



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Activation Functions

- **ReLU** (Rectified Linear Unit) – $\max(0, x)$
- **Sigmoid** – Converts to probability (0 to 1)
- **Tanh** – Scales between -1 and 1
- **Softmax** – Used in multi-class classification



```
import tensorflow.keras.backend  
as K  
def relu(x):  
    return K.maximum(0, x)  
def sigmoid(x):  
    return 1 / (1 + K.exp(-x))
```



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Training a DL Model

Train & Evaluate a Neural Network



```
history = model.fit(X_train,  
y_train, epochs=50,  
batch_size=32, validation_data=  
(X_test, y_test))  
loss, accuracy =  
model.evaluate(X_test, y_test)  
print(f"Test Accuracy: {accuracy  
* 100:.2f}%")
```



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Convolutional Neural Networks (CNNs)

Best for Image Classification & Object Detection



```
from tensorflow.keras.layers import Conv2D,
MaxPooling2D, Flatten

cnn_model = Sequential([
    Conv2D(32, (3,3), activation='relu', input_shape=
(128,128,3)),
    MaxPooling2D(2,2),
    Conv2D(64, (3,3), activation='relu'),
    MaxPooling2D(2,2),
    Flatten(),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

cnn_model.compile(optimizer='adam',
loss='categorical_crossentropy', metrics=['accuracy'])
```



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RNNs & LSTMs

Best for Sequential Data like
Time Series & NLP



```
from tensorflow.keras.layers import  
SimpleRNN, LSTM
```

```
rnn_model = Sequential([  
    LSTM(50, return_sequences=True,  
input_shape=(100, 1)),  
    LSTM(50),  
    Dense(1)  
])
```

```
rnn_model.compile(optimizer='adam',  
loss='mse')
```



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Transformers (BERT, GPT, ViTs)

Best for NLP & Vision Tasks



```
from transformers import  
TFAutoModel  
  
bert_model =  
TFAutoModel.from_pretraine  
d("bert-base-uncased")
```



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Model Regularization

Prevent Overfitting – Dropout,
L1/L2 Regularization



```
from  
tensorflow.keras.layers  
import Dropout  
  
model.add(Dropout(0.3))  
# Dropout 30%
```



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Hyperparameter Tuning

Optimize Model Performance



```
from keras_tuner import  
RandomSearch  
  
tuner = RandomSearch(model,  
objective='val_accuracy',  
max_trials=5)  
tuner.search(X_train, y_train,  
epochs=10, validation_data=  
(X_test, y_test))
```



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Deploying DL Model

Save & Load a Model



```
model.save("my_model.h5")
```

```
loaded_model =  
tf.keras.models.load_model("my_model.h5")
```



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